



Shore

**Year 11
Mathematics
Yearly Examination
September 2014**

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board approved calculators may be used
- Answer Questions 1 – 10 on the Multiple Choice answer sheet provided
- Start each of Questions 11 – 14 in a new writing booklet
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A”

Examination Number:

Set:

Total marks – 70

Section I Pages 3 - 7

10 marks

- Attempt Questions 1 – 10
- Each question is worth 1 mark
- Allow about 10 minutes for this section

Section II Pages 8 – 12

60 marks

- Attempt Questions 11 - 14
- Each question is worth 15 marks
- Allow about 1 hour and 50 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

Section I

10 marks

Attempt Questions 1 - 10

Allow about 10 minutes for this section

Use the multiple choice answer sheet.

1 If $x = 3y^3 - 2$, what is the value of y when $x = 19$?

- (A) $\sqrt[3]{\frac{17}{3}}$
- (B) $\sqrt[3]{7}$
- (C) $\sqrt[3]{21}$
- (D) $\frac{\sqrt[3]{21}}{3}$

2 What is $\frac{\sqrt{3}}{5+2\sqrt{3}}$ as a fraction with a rational denominator?

- (A) $\frac{6+5\sqrt{3}}{37}$
- (B) $\frac{5\sqrt{3}-6}{37}$
- (C) $\frac{6+5\sqrt{3}}{13}$
- (D) $\frac{5\sqrt{3}-6}{13}$

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

3 What is the domain and range of the function $y = \sqrt{4 - x^2}$?

- (A) Domain $-2 \leq x \leq 2$, Range $0 \leq y \leq 2$
- (B) Domain $-2 \leq x \leq 2$, Range $-2 \leq y \leq 2$
- (C) Domain $0 \leq x \leq 2$, Range $-4 \leq y \leq 4$
- (D) Domain $0 \leq x \leq 2$, Range $0 \leq y \leq 4$

4 What is the gradient of the normal to the curve $y = \frac{3x^2 + 2}{x^2}$ at $x = 2$?

- (A) -2
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{2}$
- (D) 2

5 What is the solution of $2^{x+1} = \frac{1}{64}$?

- (A) $x = -7$
- (B) $x = -5$
- (C) $x = 5$
- (D) $x = 7$

6 What is the derivative of $(4x^2 - 5)^3$?

- (A) $3(4x^2 - 5)$
- (B) $3(4x^2 - 5)^2$
- (C) $24x(4x^2 - 5)^2$
- (D) $12x^2(4x^2 - 5)^2$

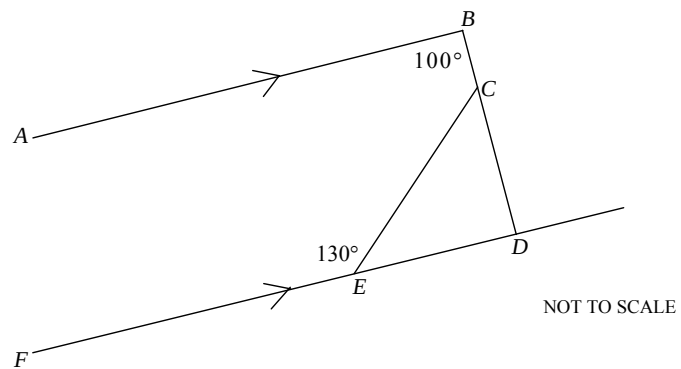
7 What is the simplified expression for $\frac{x^3 - 1}{x^2 - 1} \times \frac{x^2 - 4x - 5}{4x^2 + 4x + 4}$?

- (A) $\frac{x - 5}{4}$
- (B) $\frac{x - 1}{4}$
- (C) $\frac{x + 1}{4}$
- (D) $\frac{x^2 + x + 1}{4}$

8 What is the solution of the inequality $|3 - 2x| < 5$?

- (A) $x < -4$ or $x > 1$
- (B) $-4 < x < 1$
- (C) $-1 < x < 4$
- (D) $x < -1$ or $x > 4$

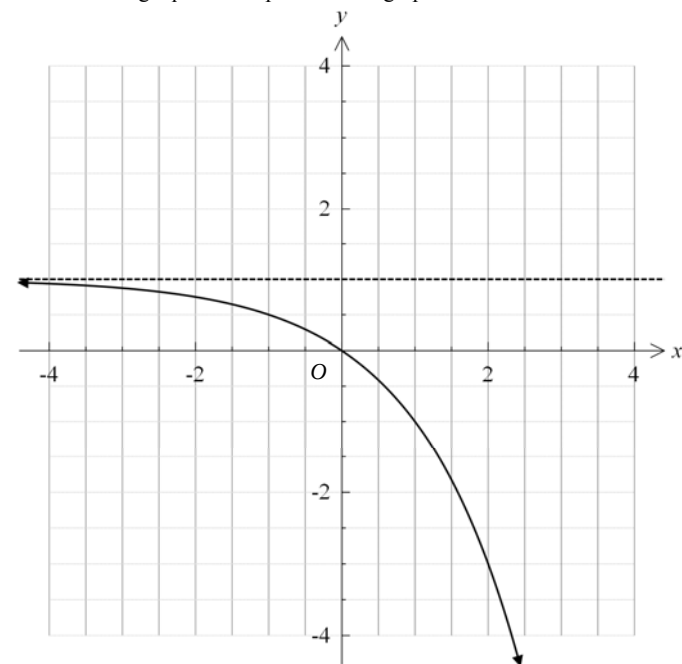
- 9 In the diagram below, AB is parallel to FD , $\angle ABC = 100^\circ$ and $\angle CEF = 130^\circ$.



What is the value of $\angle BCE$?

- (A) 100°
- (B) 110°
- (C) 120°
- (D) 130°

- 10 Which of the following equations represents this graph?



- (A) $y = 2^x - 1$
- (B) $y = 2^{-x} - 1$
- (C) $y = 1 - 2^x$
- (D) $y = 1 - 2^{-x}$

End of Section I

Section II

Total Marks 60

Attempt Questions 11 - 14

Allow about 1 hour and 50 minutes for this section.

Answer all questions, starting each question in a **new answer booklet** with your exam number clearly visible. Extra writing booklets are available.

In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

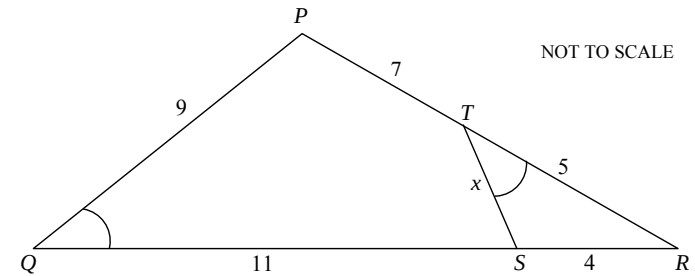
Question 11 (15 marks)

- (a) A retailer marked up the wholesale price of a jacket by 37% before selling it for \$347.98. Calculate the wholesale price of the jacket to the nearest cent. 2
- (b) Solve $\frac{x-1}{3} - 1 = \frac{x+2}{2}$. 2
- (c) What angle does the tangent to the curve $y = x^3 + x^2$ at $x = -1$ make with the positive direction of the x -axis? 2
- (d) Find the size of each interior angle of a regular octagon. 2
- (e) Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x}{3x - 5x^2}$. 2
- (f) Factorise fully $x^4 - 16$. 2

Question 11 continues on the following page

Question 11 (continued)

(g)



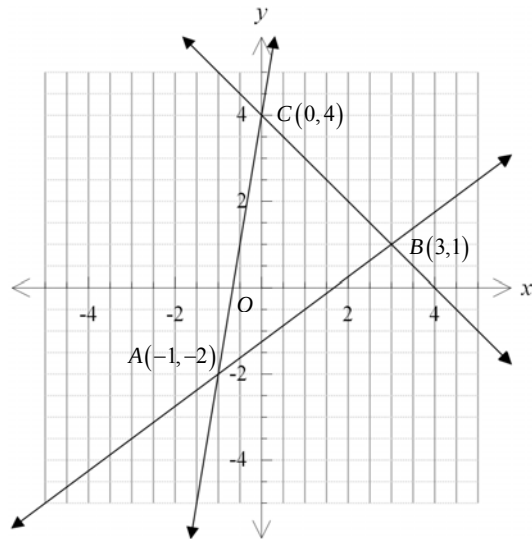
In the diagram, $\angle PQR = \angle STR$.

- (i) Prove that $\triangle PQR$ is similar to $\triangle STR$. 2
- (ii) Hence find the value of x . 1

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Differentiate $3x\sqrt{x}$. 2
- (b) If $y = x^3 - 4x^2 - 5$, for what values of x does $\frac{dy}{dx} = -4$? 3
- (c) A triangle has sides 7 cm, 9 cm and 10 cm.
- (i) Find the largest angle of the triangle. Answer to the nearest minute. 2
- (ii) Hence, or otherwise, find the area of the triangle correct to two significant figures. 1
- (d) The diagram below shows the points $A(-1, -2)$, $B(3, 1)$ and $C(0, 4)$.



- (i) Find the centre and radius of the circle which has AB as diameter. 2
- (ii) Show that the equation of the line through A and B is $3x - 4y - 5 = 0$. 2
- (iii) Find the perpendicular distance from C to the line AB . 2
- (iv) Hence find the area of the triangle ABC . 1

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) If α and β are the roots of $2x^2 - 3x - 4 = 0$, find
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $(\alpha - \beta)^2$ 2
- (b) Solve $2\sin 2\theta + \sqrt{3} = 0$ for $0^\circ \leq \theta \leq 360^\circ$. 3
- (c) Solve $|x - 3| = 3x + 1$. 3
- (d) Find the values of k for which $x^2 + kx + 4k = 0$ has real roots. 3
- (e) Prove that $\frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} = \sec^2 \theta \operatorname{cosec}^2 \theta$. 2

Question 14 (15 marks) Use a SEPARATE writing booklet

(a) Find the equation of the tangent to the curve $y = (3x + 1)(x - 2)$ which is parallel to the line $7x - y - 5 = 0$. **3**

(b) Differentiate $f(x) = x^2 - 3x$ from first principles. **3**

(c) If $y = x(2x - 1)^3$, find $\frac{dy}{dx}$ in fully factored form. **3**

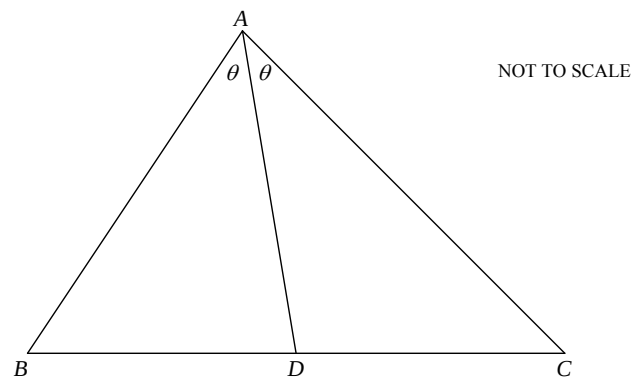
(d) Given the function $y = \frac{x^2}{x + 1}$.

(i) Find $\frac{dy}{dx}$. **2**

(ii) Find the values of x for which the tangents to this curve are horizontal. **1**

(e) In triangle ABC , AD bisects $\angle BAC$. **3**

Use the sine rule to prove that $\frac{AB}{AC} = \frac{BD}{DC}$.

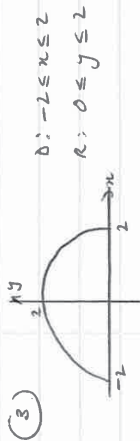


End of Examination

YEAR 11 MATHEMATICS WORKSHEET SOLUTIONS

(1) $x = 3y^2 - 2$
 $3y^2 = x + 2$
 $y^2 = \frac{x+2}{3}$
 $y = \sqrt{\frac{x+2}{3}}$
 $x=14 \rightarrow y = \sqrt{\frac{14+2}{3}} = \sqrt{6}$

(2) $\frac{\sqrt{3}}{5+2\sqrt{3}} = \frac{\sqrt{3}}{5+2\sqrt{3}} \times \frac{5-2\sqrt{3}}{5-2\sqrt{3}}$
 $= \frac{5\sqrt{3} - 2 \times 3}{25 - 4 \times 3}$
 $= \frac{5\sqrt{3} - 6}{13}$



(4) $y = 3 + \frac{2}{x^2} = 3 + 2x^{-2}$
 $\frac{dy}{dx} = -4x^{-3} = -\frac{4}{x^3}$
 When $x=2$, $m_t = \frac{dy}{dx} = -\frac{4}{8} = -\frac{1}{2}$
 $\therefore m_n = 2$

(5) $2^{z+1} = 2^{-6}$
 $\therefore z+1 = -6$
 $z = -7$

(6) $y = (4x^2 - 5)^3$
 $\therefore y' = 3(4x^2 - 5)^2 \cdot 8x = 24x(4x^2 - 5)^2$

(7) $\frac{(x-1)(x^2+x+1)}{(x-1)(x^2+x+1)} \times \frac{(x-5)(x^2+x+1)}{4(x^2+x+1)}$
 $= \frac{x-5}{4}$

(8) $|3-2x| < 5$
 $-5 < 3-2x < 5$
 $-8 < -2x < 2$
 $4 > x > -1 \therefore -1 < x < 4$

(9) $\hat{A}BC = 180^\circ - 100^\circ$ (angle L, supplementary)
 $= 80^\circ$
 $\hat{A}CB + 80^\circ = 180^\circ$ (exterior angle theorem)
 $\hat{A}CB = 100^\circ$
 $\hat{A}CB = 180^\circ - 80^\circ$ (straight angle)
 $= 100^\circ$

(10) $y = 1 - 2^x$

(11)

e) Let x be whole price
 $\therefore 137\% \text{ of } x = 347.98$
 $1.37x = 347.98$
 $x = \frac{347.98}{1.37}$

$= 254$
 $\therefore \text{whole price} = \254

f) $\frac{x-1}{3} - 1 = \frac{x+2}{4}$
 $3(x-1) - 6 = 3(x+2)$
 $3x-3-6 = 3x+6$
 $-9 = 9$
 $x = -1$

g) i) ΔPQR is a triangle
 $\hat{P}QR = \hat{R}PQ$ (common $\angle$)
 $\frac{QR}{PR} = \frac{PR}{QR} = 3$
 $\frac{PR}{QR} = \frac{12}{4} = 3$

ii) $\Delta PQR \parallel \Delta STU$ (one angle, side about $\angle$ proportion)
 $\hat{P}QR = \hat{R}PQ$ (given)
 $\hat{R}PQ = \hat{R}PQ$ (same $\angle$)
 $\therefore \Delta PQR \parallel \Delta STU$ (SAS congruence)

e) $\ln \frac{3x-2x}{x-5x^2} = \ln \frac{3-x}{x-5}$
 $= \ln \frac{3-x}{x-5}$

f) $x^4 - 16 = (x^2 - 4)(x^2 + 4)$
 $= (x-2)(x+2)(x^2 + 4)$

g) i) ΔPQR is a triangle
 $\hat{P}QR = \hat{R}PQ$ (common $\angle$)
 $\frac{QR}{PR} = \frac{PR}{QR} = 3$
 $\frac{PR}{QR} = \frac{12}{4} = 3$

ii) $\Delta PQR \parallel \Delta STU$ (one angle, side about $\angle$ proportion)

iii) $\Delta PQR \parallel \Delta STU$ (SAS congruence)
 $\hat{P}QR = \hat{R}PQ$ (given)
 $\hat{R}PQ = \hat{R}PQ$ (same $\angle$)
 $\therefore \Delta PQR \parallel \Delta STU$ (SAS congruence)

iv) $\frac{x}{4} = \frac{4}{12}$ (crossing sides proportion)
 $\therefore x = \frac{36}{12} = 3$

(12) a) $y = 3x\sqrt{x} = 3x \cdot x^{\frac{1}{2}} = 3x^{\frac{3}{2}}$ (1)
 $\frac{dy}{dx} = \frac{9}{2} x^{\frac{1}{2}} = \frac{9}{2} \sqrt{x} \text{ or } \frac{9\sqrt{x}}{2}$ (1)

b) $y = x^3 - 4x^2 - 5$

$\frac{dy}{dx} = 3x^2 - 8x$ (1)

$\frac{dy}{dx} = -4 \rightarrow 3x^2 - 8x = -4$

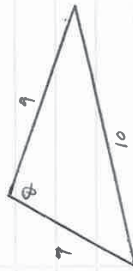
$3x^2 - 8x + 4 = 0$

$(3x-2)(x-2) = 0$ (1)

$\therefore 3x-2=0 \text{ or } x-2=0$

$x = \frac{2}{3}, 2$ (1)

c) i) largest angle is opposite largest side (1)



$\cos \theta = \frac{7^2 + 8^2 - 10^2}{2 \cdot 7 \cdot 8}$

$= \frac{0.236095238...}{1}$ (1)

$\therefore \theta = 76^\circ 14'$ (1)

ii) $A = \frac{1}{2} ab \sin C$
 $= \frac{1}{2} \cdot 7 \cdot 8 \cdot \sin 77^\circ 31'$
 $= 30.785...$
 $= 31 \text{ cm}^2$ (2, f)

(12) i) $AB = \sqrt{(3-1)^2 + (1-2)^2}$
 $= \sqrt{4+1}$
 $= \sqrt{5}$ (1)

map of AB: $x = \frac{x_1+y_1}{2}$ $y = \frac{y_1+y_2}{2}$

$= \frac{-1+3}{2}$ $= \frac{-2+1}{2}$

$= 1$ $= -\frac{1}{2}$ (1)

$\therefore$ Centre is $(1, -\frac{1}{2})$, radius $= \frac{5}{2}$

ii) $\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$

$\frac{y+1}{x+1} = \frac{3}{4}$ (1)

$4y+4 = 3x+3$

$\therefore 3x-4y-5=0$ (1)

iii) $r = \frac{|Ax_1+By_1+C|}{\sqrt{A^2+B^2}}$
 $= \frac{|3(0)-4(4)-5|}{\sqrt{9+16}}$ (1)

$= \frac{|-21|}{5}$

$= \frac{21}{5}$ (1)

iv) $Area = \frac{1}{2} ab \sin C$
 $= \frac{1}{2} \cdot 5 \times \frac{21}{5}$
 $= \frac{21}{2}$
 $= 10\frac{1}{2}$ (1)

(13) (a) $2x^2 - 3x - 4 = 0$
 i) $x+p = -\frac{b}{a} = \frac{3}{2}$ (1)
 ii) $x+p = \frac{c}{a} = -\frac{4}{2} = -2$ (1)

iii) $(x-p)^2 = x^2 - 2xp + p^2$
 $= (x^2 + p^2) - 2xp$
 $= (x+p)^2 - 4xp$ (1)
 $= \frac{9}{4} - 4 \cdot -2$
 $= 10\frac{1}{4}$ (1)

(b) $\sin \theta = \frac{1}{\sqrt{10}} + \frac{1}{\sqrt{10}}$
 $= \frac{\sqrt{10} + \sqrt{10}}{10}$

$= \frac{2}{\sqrt{10}}$ (1)

$= \frac{1}{\sqrt{10}} \cdot \frac{1}{\sqrt{10}}$

$= \frac{1}{10}$ (1)

$= \frac{1}{10}$ (1)

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(10) a) $y = (3x+1)(x-2) = 3x^2 - 5x - 2$

$\frac{dy}{dx} = 6x - 5$ (1)

Since if given line $= -\frac{7}{1} = 7$ (1)

We require $\frac{dy}{dx} = 7$

$6x - 5 = 7$

$6x = 12$

$x = 2$

When $x = 2, y = 7 \times 0 = 0$ (1)

Eg- if eqn: $(2, 0), m = 7$

$y - y_1 = m(x - x_1)$

$y - 0 = 7(x - 2)$

$y = 7x - 14$ (1)

(11) $f(x) = x^2 - 3x$

$f(x+h) = (x+h)^2 - 3(x+h)$

$= x^2 + 2xh + h^2 - 3x - 3h$

$f(x+h) - f(x) = x^2 + 2xh + h^2 - 3x - 3h - (x^2 - 3x)$

$= 2xh + h^2 - 3h$ (1)

$\frac{f(x+h) - f(x)}{h} = \frac{2xh + h^2 - 3h}{h}$

$= 2x + h - 3$ (1)

$f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right]$

$= \lim_{h \rightarrow 0} [2x + h - 3]$

$= 2x - 3$ (1)

(12) $y = x(2x-1)^3$

$y' = x \cdot 3(2x-1)^2 \cdot 2 + (2x-1)^3$ (1)

$= 6x(2x-1)^2 + (2x-1)^3$

$= (2x-1)^2 [6x + 2x-1]$ (1)

$= (2x-1)^2 (8x-1)$ (1)

(13) $y = \frac{x^2}{x+1}$

(i) $\frac{dy}{dx} = \frac{(x+1) \cdot 2x - x^2 \cdot 1}{(x+1)^2}$ (1)

$= \frac{2x^2 + 2x - x^2}{(x+1)^2}$

$= \frac{x^2 + 2x}{(x+1)^2}$ (1)

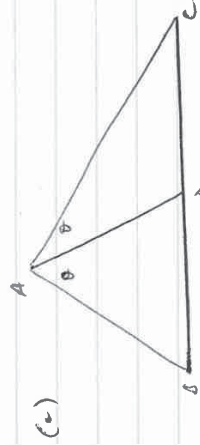
(ii) we require $\frac{dy}{dx} = 0$

$\therefore \frac{x^2 + 2x}{(x+1)^2} = 0$

$\therefore x^2 + 2x = 0$

$x(x+2) = 0$ (1)

$\therefore x = -2, 0$



Let $\angle ABD = x$ $\therefore \angle ADC = 180^\circ - x$

In $\triangle ABD$, $\frac{BD}{AB} = \frac{\sin B}{\sin \alpha}$

In $\triangle ADC$, $\frac{DC}{AC} = \frac{\sin B}{\sin(180^\circ - x)} = \frac{\sin B}{\sin x}$

$\therefore \frac{BD}{AB} = \frac{DC}{AC}$

$\therefore BD \cdot AC = DC \cdot AB$

$\frac{BD}{DC} = \frac{AB}{AC}$

(1) for sine rule rule

(1) for $\sin(180^\circ - \theta) = \sin \theta$

(1) correct ratios

Q13 In $\triangle ABD$, $\frac{\sin B}{BD} = \frac{\sin C}{AD}$

$\therefore \sin B = \frac{AD \sin C}{BD}$ ---- (1)

In $\triangle ADC$, $\frac{\sin B}{DC} = \frac{\sin C}{AD}$

$\therefore \sin C = \frac{AD \sin B}{DC}$ ---- (2)

In $\triangle ABC$, $\frac{\sin B}{AC} = \frac{\sin C}{AB}$

$\therefore \frac{\sin C}{\sin B} = \frac{AB}{AC}$

From (1) and (2), $\frac{AB}{AC} = \frac{AD \sin B}{DC} \cdot \frac{AD \sin C}{BD}$

$= \frac{BD}{DC}$

(1) for sine rule rule

(2) for sine rule another two times

(1) for correct ratios